

Partially symmetric Macdonald polynomials and AFLT type constant term identities

Zihao Huang

Central South University

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(a) 蒋文龙



(b) 周岳

What is the Macdonald polynomials?

Symmetric function ring

- ▶ In what follows, we fix $\mathbb{F} = \mathbb{Q}(q, t)$ and let $\mathfrak{S}_n$ be the symmetric group on $\{1, \dots, n\}$.
- ▶ We say $f \in \mathbb{F}[x_1, \dots, x_n]$ is a **symmetric function**, if

$$\sigma(f) := f(x_{\sigma(1)}, \dots, x_{\sigma(n)}) = f(x_1, \dots, x_n) \quad \text{for all } \sigma \in \mathfrak{S}_n.$$

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- ▶ A standard basis for the space of symmetric functions is the set of **monomial symmetric functions**: let $x := (x_1, \dots, x_n)$, for an integer partition λ , define

$$m_\lambda(x) := \sum_{\alpha \in \mathbb{N}^n, \alpha^+ = \lambda} x_1^{\alpha_1} \cdots x_n^{\alpha_n},$$

where α^+ is the rearrangement of α in weakly decreasing order.

Macdonald polynomials

Define a scalar product on $\mathbb{F}[x_1, \dots, x_n]$ by

$$\langle f, g \rangle_n = \langle f, g \rangle_{q,t,n} := \text{CT}_x \left(f(x; q, t) g(x^{-1}; q^{-1}, t^{-1}) W(x) \right),$$

where

$$W(x) = W(x; q, t) := \prod_{1 \leq i < j \leq n} \frac{(x_i/x_j)_\infty (qx_i/x_j)_\infty}{(tx_i/x_j)_\infty (qtx_i/x_j)_\infty}$$

is the weight function.

Macdonald polynomials

The symmetric Macdonald polynomials $P_\lambda(x; q, t)$ are the unique symmetric functions such that

- ▶ $\langle P_\lambda(x; q, t), P_\mu(x; q, t) \rangle_n = 0$ if $\lambda \neq \mu$
- ▶ $P_\lambda(x; q, t) = m_\lambda + \sum_{\mu < \lambda} c_{\lambda\mu} m_\mu, \quad c_{\lambda\mu} \in \mathbb{F},$

with respect to the dominance order on partitions.

The nonsymmetric Macdonald polynomials $E_\nu(x; q, t) \in \mathbb{F}[x_1, \dots, x_n]$, $\nu \in \mathbb{N}^n$ are the unique polynomials such that

- ▶ $\langle E_\nu(x; q, t), E_\eta(x; q, t) \rangle_n = 0$ if $\nu \neq \eta$
- ▶ $E_\nu(x; q, t) = x^\nu + \sum_{\eta \prec \nu} c_{\nu\eta} x^\eta, \quad c_{\nu\eta} \in \mathbb{F},$

with respect to the Bruhat order on compositions.

Symmetric functions, especially the Macdonald polynomials, and constant term identities are deeply associated with each other. Their connection appears in:

- ▶ Macdonald constant term conjecture;
- ▶ Dyson orthogonality problem;
- ▶ Selberg type integral;

⋮

The Selberg integral (1944)

In 1944, **Atle Selberg** gave a remarkable n dimensional generalization of the Euler Beta integral:

$$\int_0^1 \cdots \int_0^1 \prod_{i=1}^n x_i^{\alpha-1} (1-x_i)^{\beta-1} \prod_{1 \leq i < j \leq n} (x_i - x_j)^{2\gamma} dx_1 \cdots dx_n \\ = \prod_{j=0}^{n-1} \frac{\Gamma(\alpha + j\gamma) \Gamma(\beta + j\gamma) \Gamma(1 + (j+1)\gamma)}{\Gamma(\alpha + \beta + (n+j-1)\gamma) \Gamma(1 + \gamma)},$$

where α, β, γ are complex parameters such that $\operatorname{Re}(\alpha) > 0$, $\operatorname{Re}(\beta) > 0$, $\operatorname{Re}(\gamma) > -\min\{1/n, \operatorname{Re}(\alpha)/(n-1), \operatorname{Re}(\beta)/(n-1)\}$.



A. Selberg, *Bemerkninger om et multipelt integral (Remarks on a multiple integral)*, Norsk Mat. Tidsskr. 26 (1944), 71–78.

The Selberg integral

The **Selberg integral** has appeared in random matrix theory, analytic number theory, enumerative combinatorics, conformal field theory, . . .



Figure: Atle Selberg

It is widely regarded as one of the most fundamental and important hypergeometric integrals.



P. J. Forrester, S. O. Warnaar, *The importance of the Selberg integral*, Bull. Am. Math. Soc. 45 (2008) 489–534.

q -Selberg integral

In 1980, **Askey** conjectured the next q -analogue of the Selberg integral:

$$\begin{aligned} & \int_{[0,1]^n} \prod_{i=1}^n x_i^{\alpha-1} (qx_i)^{\beta-1} \prod_{1 \leq i < j \leq n} x_j^{2\gamma} (q^{1-\gamma} x_i/x_j)^{2\gamma} d_q x_1 \cdots d_q x_n \\ &= q^{\alpha\gamma\binom{n}{2} + 2\gamma^2\binom{n}{3}} \prod_{j=0}^{n-1} \frac{\Gamma_q(\alpha + \gamma(j-1))\Gamma_q(\beta + \gamma(j-1))\Gamma_q(1 + \gamma j)}{\Gamma_q(\alpha + \beta + \gamma(n+j-2))\Gamma_q(1 + \gamma)}, \end{aligned}$$

Here

$$(a)_z = (a; q)_z := \prod_{j=0}^{\infty} \frac{1 - aq^j}{1 - aq^{z+j}}$$

is the q -shifted factorial, $\int_{[0,1]^n} d_q x$ is the q -integral, and $\Gamma_q(x)$ is the q -Gamma function.



R. A. Askey, *Some basic hypergeometric extensions of integrals of Selberg and Andrews*, SIAM J. Math. Anal. (1980).

Equivalence to a constant term identity

This conjecture was proved independently by **Habsieger** and **Kadell**, who also turned the q -integral into an equivalent constant term identity:

$$\int_{[0,1]^n} \prod_{i=1}^n x_i^{\alpha-1} (qx_i)_{\beta-1} \prod_{1 \leq i < j \leq n} x_j^{2\gamma} (q^{1-\gamma} x_i/x_j)_{2\gamma} d_q x$$

$$= q^{\alpha\gamma} \binom{n}{2} + 2\gamma^2 \binom{n}{3} \prod_{j=0}^{n-1} \frac{\Gamma_q(\alpha + \gamma(j-1)) \Gamma_q(\beta + \gamma(j-1)) \Gamma_q(1 + \gamma j)}{\Gamma_q(\alpha + \beta + \gamma(n+j-2)) \Gamma_q(1 + \gamma)},$$

$\Updownarrow$

$$\text{CT}_x \prod_{i=1}^n (1/x_i)_a (qx_i)_b \prod_{1 \leq i < j \leq n} (x_i/x_j)_c (qx_j/x_i)_c = \prod_{i=1}^n \frac{(q^{a+(i-1)c+1})_b (q)_{ic}}{(q)_{b+(i-1)c} (q)_c},$$

where $a, b, c \in \mathbb{N}$. Here CT_x means taking the constant term with respect to x .



L. Habsieger, *Une q -intégrale de Selberg–Askey*, SIAM J. Math. Anal.(1988).



K. W. J. Kadell, *A proof of Askey's conjectured q -analogue of Selberg's integral and a conjecture of Morris*, SIAM J. Math. Anal.(1988).

The q -Morris identity

The identity:

$$\text{CT}_x \prod_{i=1}^n (1/x_i)_a (qx_i)_b \prod_{1 \leq i < j \leq n} (x_i/x_j)_c (qx_j/x_i)_c = \prod_{i=1}^n \frac{(q^{a+(i-1)c+1})_b (q)_{ic}}{(q)_{b+(i-1)c} (q)_c}$$

and its equivalence to the q -Selberg integral were first conjectured by **Morris** in his Ph.D. thesis. For this reason, the above identity is now commonly referred to as the Habsieger–Kadell q -Morris identity, or **q -Morris identity** for short.



W. G. Morris, *Constant Term Identities for Finite and Affine Root System: Conjectures and Theorems*, Ph.D. thesis, University of Wisconsin–Madison, 1982.

The Kadell's integral

Let λ be an integer partition and $P_\lambda^{(1/\gamma)}(x)$ be the Jack polynomial. Then

$$\int_{[0,1]^k} P_\lambda^{(1/\gamma)}(x) \prod_{i=1}^k x_i^{\alpha-1} (1-x_i)^{\beta-1} \prod_{1 \leq i < j \leq k} (x_i - x_j)^{2\gamma} dx \\ = P_\lambda^{(1/\gamma)}(1, \dots, 1) \prod_{i=1}^k \frac{\Gamma(\alpha + (k-i)\gamma + \lambda_i) \Gamma(\beta + (i-1)\gamma) \Gamma(1+i\gamma)}{\Gamma(\alpha + \beta + (2k-i-1)\gamma + \lambda_i) \Gamma(1+\gamma)},$$

where $\operatorname{Re}(\alpha) > -\lambda_k$, $\operatorname{Re}(\beta) > 0$, $\operatorname{Re}(\gamma) > \dots$. The above identity was proved by Kadell in 1987; at the time he didn't notice that $P_\lambda^{(1/\gamma)}$ is actually the Jack polynomial, so he called such a polynomial the "Selberg polynomial".



K.W.J. Kadell, *The Selberg–Jack symmetric functions*, Adv. Math. 130 (1997) 33–102.

Kadell's Problem

For a few partitions of small size Kadell was able to show that a q -analogue of the Jack symmetric functions existed which featured in a q -analogue of his extension of Selberg integral. He posed the problem:

"finding q -analogues of these polynomials for all partitions".

$$\int_{[0,1]^k} P_{\lambda}^{(1/\gamma)}(x) \prod_{i=1}^k x_i^{\alpha-1} (1-x_i)^{\beta-1} \prod_{1 \leq i < j \leq k} (x_i - x_j)^{2\gamma} dx = \cdots$$

$\Downarrow$

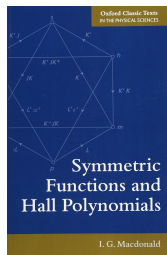
$$\int_{[0,1]^n} ? \times \prod_{i=1}^n x_i^{\alpha-1} (qx_i)_{\beta-1} \prod_{1 \leq i < j \leq n} x_j^{2\gamma} (q^{1-\gamma} x_i/x_j)_{2\gamma} d_q x = \cdots$$

The Macdonald integral

The answer to Kadell's problem was soon given by Macdonald in his book.

$$\int_{(0,1)^n} P_\lambda(x; q, t) \prod_{i=1}^n x_i^{\alpha-1} (qx_i)_{\beta-1} \prod_{1 \leq i < j \leq n} x_j^{2\gamma} (q^{1-\gamma} x_i/x_j)_{2\gamma} d_q x$$

$$= q^F \prod_{i=1}^n \frac{\Gamma_q(\lambda_i + a + (i-1)k) \Gamma_q(b + (i-1)k)}{\Gamma_q(\lambda_i + a + b + (n+i-2)k)} \prod_{1 \leq i < j \leq n} \frac{\Gamma_q(\lambda_i - \lambda_j + (j-i+1)k)}{\Gamma_q(\lambda_i - \lambda_j + (j-i)k)}$$



What problem do we care about?

Nonsymmetric generalization

In 1995, Baker and Forrester conjectured that: for nonnegative integers a, b, c and positive integers $m \leq n$, there holds

$$\begin{aligned} & \text{CT}_x \prod_{i=1}^n (x_0/x_i)_a (qx_i/x_0)_b \prod_{1 \leq i < j \leq n} (x_i/x_j)_{c+\chi(i,j \leq m)} (qx_j/x_i)_{c+\chi(i,j \leq m)} \\ &= \prod_{i=0}^{n-1} \frac{(q^{ic+\chi(i \geq n-m)(i-n+m)+a+1})_b (q)_{(i+1)c+\chi(i > n-m)(i-n+m)}}{(q)_{ic+\chi(i \geq n-m)(i-n+m)+b} (q)_{c+\chi(i > n-m)}} \\ & \quad \times \prod_{i=1}^{m-1} (1 - q^{(i+1)(c+1)}). \end{aligned}$$

This conjecture was proved in 2015 by Károlyi et.al.



T. H. Baker and P. J. Forrester, *Generalizations of the q -Morris constant term identity*, J. Comb. Theory A (1998).



G. Károlyi, Z. L. Nagy, F. V. Petrov, and V. Volkov, *A new approach to constant term identities and Selberg-type integrals*, Adv. Math. (2015).

Main Question:

Kadell:

$$\text{CT}_x P_\lambda^{(1/\gamma)}(x^{-1}) \prod_{i=1}^n (1 - 1/x_i)^a (1 - x_i)^b \prod_{1 \leq i \neq j \leq n} (1 - x_i/x_j)^c = \dots$$

$\Downarrow$

Macdonald:

$$\text{CT}_x P_\lambda(x^{-1}; q, q^c) \prod_{i=1}^n (1/x_i)_a (qx_i)_b \prod_{1 \leq i < j \leq n} (x_i/x_j)_c (qx_j/x_i)_c = \dots$$

$\Downarrow$

$$\text{CT}_x ? \times \prod_{i=1}^n (1/x_i)_a (qx_i)_b \prod_{1 \leq i < j \leq n} (x_i/x_j)_{c+\chi(j \leq m)} (qx_j/x_i)_{c+\chi(j \leq m)}$$

Is there any object featured in the Baker–Forrester's constant term?

Based on computer data, we find that the next constant term has a nice product form when $\ell(\lambda) < m$.

$$\begin{aligned} \text{CT}_x P_\lambda \left(\left[X_m^{-1} + \frac{1-q^c}{1-q^{c+1}} X_{n \setminus m}^{-1} \right]; q, q^{c+1} \right) & \prod_{i=1}^n (1/x_i)_a (qx_i)_b \\ & \times \prod_{1 \leq i < j \leq n} (x_i/x_j)_{c+\chi(j \leq m)} (qx_j/x_i)_{c+\chi(j \leq m)} = \cdots, \end{aligned}$$

where

$$X_m^{-1} := x_1^{-1} + \cdots + x_m^{-1}, \quad X_{n \setminus m}^{-1} := x_{m+1}^{-1} + \cdots + x_n^{-1}.$$

$$\Downarrow$$

$$\text{CT}_x P_\lambda(x^{-1}; q, q^c) \prod_{i=1}^n (1/x_i)_a (qx_i)_b \prod_{1 \leq i < j \leq n} (x_i/x_j)_c (qx_j/x_i)_c = \cdots.$$

when $m = n, c \mapsto c - 1$.

"What is the connection between the BF-constant term and the Macdonald polynomial theory?"

An observation

$$\begin{aligned} & {}^{\text{CT}}_x P_\lambda \left(\left[X_m^{-1} + \frac{1-q^c}{1-q^{c+1}} X_{n \setminus m}^{-1} \right]; q, q^{c+1} \right) \prod_{i=1}^n (1/x_i)_a (qx_i)_b \\ & \quad \times \prod_{1 \leq i < j \leq n} (x_i/x_j)_{c+\chi(j \leq m)} (qx_j/x_i)_{c+\chi(j \leq m)} \\ & \quad \Downarrow \end{aligned}$$

$$\begin{aligned} & {}^{\text{CT}}_x P_\lambda \left(\left[X_m^{-1} + \frac{1-q^c}{1-q^{c+1}} X_{n \setminus m}^{-1} \right]; q, q^{c+1} \right) \prod_{1 \leq i < j \leq m} (q^{-c} x_i^{-1} - x_j^{-1}) \\ & \times q^{c \binom{m}{2}} \prod_{1 \leq i < j \leq m} (x_i - q^{c+1} x_j) \prod_{i=1}^n (1/x_i)_a (qx_i)_b \prod_{1 \leq i < j \leq n} (x_i/x_j)_c (qx_j/x_i)_c. \end{aligned}$$

This observation leads us to consider the **Hecke anti-symmetrization** on the variables $x_1, \dots, x_m$.

What is the Hecke symmetrization and anti-symmetrization?

The Hecke symmetrization and anti-symmetrization are t -deformation of the usual ones:

$$\mathcal{S}^{(X_n)} := \sum_{\sigma \in S_n} \sigma$$

$$\mathcal{A}^{(X_n)} := \sum_{\sigma \in S_n} (-1)^{-\ell(\sigma)} \sigma$$

- For $i = 1, \dots, n-1$, let $s_i = (i, i+1)$ be the transposition and define

$$T_i = ts_i + \frac{1-t}{1-x_i/x_{i+1}}(s_i - 1)$$

be the operator that acts on $\mathbb{F}[x_1, \dots, x_n]$.

- For $\sigma \in \mathfrak{S}_n$ with reduced word decomposition $s_{i_1} \cdots s_{i_\ell}$, define

$$T_\sigma := T_{i_1} \cdots T_{i_\ell}.$$

Note: $T_\sigma|_{t=1} = \sigma$.

$\Downarrow$

$$\mathcal{S}_t^{(X_n)} := \sum_{\sigma \in S_n} T_\sigma$$

$$\mathcal{A}_t^{(X_n)} := \sum_{\sigma \in S_n} (-t)^{-\ell(\sigma)} T_\sigma.$$

Example

let $\delta^n := (n-1, \dots, 2, 1, 0)$ be the staircase partition, we have

$$\mathcal{A}^{(X_n)}(x^\delta) = \sum_{\sigma \in S_n} (-1)^{-\ell(\sigma)} \sigma(x^{\delta^n}) = \prod_{1 \leq i < j \leq n} (x_i - x_j).$$

$\Downarrow$

$$\mathcal{A}_t^{(X_n)}(x^\delta) = \sum_{\sigma \in S_n} (-t)^{-\ell(\sigma)} T_\sigma(x^{\delta^n}) = \prod_{1 \leq i < j \leq n} (tx_i - x_j)$$

We refer $\Delta_t(X_n) = \prod_{1 \leq i < j \leq n} (tx_i - x_j)$ as the t -Vandermonde determinant.

Example

Let $\eta \in \mathbb{N}^n$ and $E_\eta(x; q, t)$ be the nonsymmetric Macdonald polynomials, then

- ▶ $\mathcal{S}_t^{(X_n)} E_\eta(x; q, t) = c_\eta P_{\eta^+}(x; q, t);$
- ▶ $\mathcal{A}_t^{(X_n)} E_{\eta+\delta^n}(x; q, t) = c'_\eta P_{\eta^+}(x; q, \textcolor{red}{qt}) \prod_{1 \leq i < j \leq n} (tx_i - x_j).$

When $t = q^c$, we have

$$\mathcal{A}_{q^c}^{(X_n)} E_{\eta+\delta^n}(x; q, q^c) = c'_\eta P_\lambda(x; q, \textcolor{red}{q}^{c+1}) \prod_{1 \leq i < j \leq n} (q^c x_i - x_j).$$

Quite familiar?

Example

Let $\eta \in \mathbb{N}^n$, $E_\eta(x; q, t)$ be the nonsymmetric Macdonald polynomials.
Then

- ▶ $\mathcal{S}_t^{(X_n)} E_\eta(x; q, t) = c_\eta P_{\eta^+}(x; q, t);$
- ▶ $\mathcal{A}_t^{(X_n)} E_{\eta+\delta^n}(x; q, t) = c'_\eta P_{\eta^+}(x; q, \textcolor{red}{t}) \Delta_t(X_n).$

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$$\mathcal{A}_{q^c}^{(X_n)} E_{\eta+\delta^n}(x; q, q^c) = c'_\eta P_\lambda(x; q, \textcolor{red}{q^{c+1}}) \prod_{1 \leq i < j \leq n} (q^c x_i - x_j).$$

What we just considered is :

$$\begin{aligned} & \text{CT}_x P_\lambda \left(\left[X_m^{-1} + \frac{1 - q^c}{1 - q^{c+1}} X_{n \setminus m}^{-1} \right]; q, q^{c+1} \right) \prod_{1 \leq i < j \leq m} (q^{-c} x_i^{-1} - x_j^{-1}) \\ & \times q^{c \binom{m}{2}} \prod_{1 \leq i < j \leq m} (x_i - q^{c+1} x_j) \prod_{i=1}^n (1/x_i)_a (q x_i)_b \prod_{1 \leq i < j \leq n} (x_i/x_j)_c (q x_j/x_i)_c. \end{aligned}$$

Question

It is now natural to conjecture that, for integer partition λ with $\ell(\lambda) < m$, there holds

$$\mathcal{A}_t^{(X_m)} E_{(\lambda + \delta^m, 0^{n-m})}(x; q, t) = c_\lambda P_\lambda \left(\left[X_m + \frac{q(1-t)}{1-qt} X_{n \setminus m} \right]; q, qt \right) \Delta_t(X_m)$$

for some $c_\lambda \in \mathbb{F}$.

Previous Work

- In 2000, **Baker–Dunkl–Forrester** proved that, for partition ξ with $\ell(\xi) \leq n - m$, $\xi_1 \leq m$, there holds:

$$\mathcal{A}_t^{(X_m)} \mathcal{S}_t^{(X_{n \setminus m})} E_{(\delta^m, \xi)}(x; q, t) = c_\xi \Delta_t(X_m) P_\xi \left(\left[X_{n \setminus m} \right]; qt, t \right).$$

- In 2015, based on their studies on Macdonald polynomials in super-space, **O. Blondeau-Fournier, L. Lapointe and P. Mathieu**, showed that, for partitions λ and ξ with $\ell(\lambda) \leq m$, $\ell(\xi) \leq n - m$,

$$\begin{aligned} & \mathcal{A}_t^{(X_m)} \mathcal{S}_t^{(X_{n \setminus m})} E_{(\lambda + \delta^m, \xi)}(x; q, t) \\ &= c_\eta \Delta_t(X_m) P_\lambda \left(\left[X_m + \frac{q(1-t)}{1-qt} X_{n \setminus m} \right]; q, qt \right) P_\xi \left(\left[X_{n \setminus m} \right]; qt, t \right) \end{aligned}$$

when $|\lambda| + |\xi| \leq \min\{m, n - m\}$.

Here:

$$\begin{aligned} \delta^m &= (m - 1, \dots, 1, 0), \\ (\delta^m, \xi) &= (m - 1, \dots, 1, 0, \xi_1, \dots, \xi_{n-m}), \\ (\lambda + \delta^m, \xi) &= (\lambda_1 + m, \dots, \lambda_{m-1} + 1, \lambda_m, \xi_1, \dots, \xi_{n-m}). \end{aligned}$$

- In 2000, **Baker–Dunkl–Forrester** proved that, for partition ξ with $\ell(\xi) \leq n - m$, $\xi_1 \leq m$, there holds:

$$\mathcal{A}_t^{(X_m)} \mathcal{S}_t^{(X_{n \setminus m})} E_{(\delta^m, \xi)}(x; q, t) = c_\xi \Delta_t(X_m) P_\xi \left(\left[X_{n \setminus m} \right]; qt, t \right).$$

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$$\begin{aligned} \mathcal{A}_t^{(X_m)} \mathcal{S}_t^{(X_{n \setminus m})} E_{(\lambda + \delta^m, \xi)}(x; q, t) \\ = c_{\lambda, \xi} \Delta_t(X_m) P_\lambda \left(\left[X_m + \frac{q(1-t)}{1-qt} X_{n \setminus m} \right]; q, qt \right) P_\xi \left(\left[X_{n \setminus m} \right]; qt, t \right) \end{aligned}$$

when $|\lambda| + |\xi| \leq \min\{m, n - m\}$.

- Recently, we prove by a constant term method that

$$\mathcal{A}_t^{(X_m)} E_{(\lambda + \delta^m, 0^{n-m})}(x; q, t) = c_\lambda \Delta_t(X_m) P_\lambda \left(\left[X_m + \frac{q(1-t)}{1-qt} X_{n \setminus m} \right]; q, qt \right)$$

holds when $\ell(\lambda) < m$.

We conclude these three results as the next theorem.

Theorem

Let n, m be positive integers such that $m \leq n$, λ, ξ be partitions such that $\ell(\lambda) < m$, $\ell(\xi) \leq n - m$, and $\eta := (\lambda + \delta^m, \xi)$. Then we can write

$$\begin{aligned} \mathcal{A}_t^{(X_m)} \mathcal{S}_t^{(X_{n \setminus m})} E_\eta(x; q, t) \\ = c_\eta \Delta_t(X_m) P_\lambda \left(\left[X_m + \frac{q(1-t)}{1-qt} X_{n \setminus m} \right]; q, qt \right) P_\xi \left([X_{n \setminus m}]; qt, t \right) \end{aligned}$$

in the following cases:

1. $\xi = \emptyset$; (what we just conjecture and proved)
2. $\lambda = \emptyset$ and $\xi_1 \leq m$; (Baker et.al.)
3. $\lambda, \xi \neq \emptyset$, and $|\lambda| + |\xi| \leq \min\{n - m, m\}$ (O. Blondeau-Fournier et.al.).

Here $c_\eta \in \mathbb{F}$ is a nonzero constant.

Application

$$\begin{aligned} & \text{CT}_x P_\lambda \left(\left[X_m^{-1} + \frac{1-q^c}{1-q^{c+1}} X_{n \setminus m}^{-1} \right]; q, q^{c+1} \right) \Delta_{q^{-c}}(X_m^{-1}) \\ & \times q^{c \binom{m}{2}} \prod_{1 \leq i < j \leq m} (x_i - q^{c+1} x_j) \prod_{i=1}^n (1/x_i)_a (qx_i)_b \prod_{1 \leq i < j \leq n} (x_i/x_j)_c (qx_j/x_i)_c. \end{aligned}$$

$\Downarrow$

$$\begin{aligned} & \sum_{\alpha \in \mathbb{N}^n} \sum_{\substack{\eta \in \mathbb{N}^n \\ \eta^+ = \lambda + \delta^m}} \text{CT}_x x^\alpha E_\eta(x^{-1}; q^{-1}, q^{-c}) \\ & \times \prod_{i=1}^n (1/x_i)_a (qx_i)_b \prod_{1 \leq i < j \leq n} (x_i/x_j)_c (qx_j/x_i)_c. \end{aligned}$$

The problem then reduces to the relation between the nonsymmetric Macdonald polynomials and the q -Morris type constant term identity.

We go further on this problem

3. The Alba-Fateev-Litvinov-Tarnopolskiy (AFLT) integral In 2011, in attempt to solve the AGT conjecture, Alba-Fateev-Litvinov-Tarnopolskiy obtain the following integral:

$$\begin{aligned} & \int_{[0,1]^k} P_\lambda^{(1/\gamma)}(x) P_\mu^{(1/\gamma)}[x + \beta/\gamma - 1] \prod_{i=1}^k x_i^{\alpha-1} (1-x_i)^{\beta-1} \prod_{1 \leq i < j \leq k} (x_i - x_j)^{2\gamma} dx \\ &= P_\lambda^{(1/\gamma)}[k] P_\mu^{(1/\gamma)}[k + \beta/\gamma - 1] \\ & \times \prod_{i=1}^k \frac{\Gamma(\alpha + (k-i)\gamma + \lambda_i) \Gamma(\beta + (i-1)\gamma) \Gamma(1+i\gamma)}{\Gamma(\alpha + \beta + (2k-l-i-1)\gamma + \lambda_i) \Gamma(1+\gamma)} \\ & \times \prod_{i=1}^k \prod_{j=1}^l \frac{\Gamma(\alpha + \beta + (2k-i-j-1)\gamma + \lambda_i + \mu_j)}{\Gamma(\alpha + \beta + (2k-i-j)\gamma + \lambda_i + \mu_j)}, \end{aligned}$$

where l is an arbitrary integer such that $l \geq \ell(\mu)$.



V.A. Alba, V.A. Fateev, A.V. Litvinov, G.M. Tarnopolskiy, *On combinatorial expansion of the conformal blocks arising from AGT conjecture*, Lett. Math. Phys. 98 (2011) 33–64.

4. **AFLT-type q -Selberg integral** For $\lambda \in \mathcal{P}_n$, $\mu \in \mathcal{P}$, and $a, b, q, t \in \mathbb{C}$ such that $|b|, |q|, |t| < 1$, there holds

$$\begin{aligned} & \frac{1}{n!(2\pi i)^n} \int_{\mathbb{T}^n} P_\lambda(z; q, t) P_\mu \left(\left[z + \frac{t-b}{1-t} \right]; q, t \right) \prod_{i=1}^n \frac{(a/z_i, qz_i/a; q)_\infty}{(b/z_i, z_i; q)_\infty} \\ & \quad \times \prod_{1 \leq i < j \leq n} \frac{(z_i/z_j, z_j/z_i; q)_\infty}{(tz_i/z_j, tz_j/z_i; q)_\infty} \frac{dz_1}{z_1} \cdots \frac{dz_n}{z_n} \\ & = b^{|\lambda|} t^{|\mu|} P_\lambda \left(\left[\frac{1-t^n}{1-t} \right]; q, t \right) P_\mu \left(\left[\frac{1-bt^{n-1}}{1-t} \right]; q, t \right) \\ & \quad \times \prod_{i=1}^n \frac{(t, at^{n-m-i} q^{\lambda_i}, at^{1-i}/b, qt^{i-1}b/a; q)_\infty}{(q, t^i, bt^{i-1}, at^{1-i} q^{\lambda_i}/b; q)_\infty} \\ & \quad \times \prod_{i=1}^n \prod_{j=1}^m \frac{(at^{n-i-j+1} q^{\lambda_i + \mu_j}; q)_\infty}{(at^{n-i-j} q^{\lambda_i + \mu_j}; q)_\infty}, \end{aligned}$$

where m is an arbitrary integer such that $m \geq \ell(\mu)$, and $P_\lambda(x; q, t)$ is the Macdonald polynomial.



S. P. Albion, E. M. Rains and S. O. Warnaar, *AFLT-type Selberg integrals*, Commun. Math. Phys. 388 (2021), 735–791.

AFLT-type q -Morris identity

By the standard transformation between q -Selberg type integrals and q -Morris type constant term identities, one can rewrite the AFLT-type q -Selberg integral to an equivalent constant term identity.



W. G. Morris, *Constant Term Identities for Finite and Affine Root System: Conjectures and Theorems*, Ph.D. thesis, University of Wisconsin–Madison, 1982.

AFLT-type q -Morris identity

For nonnegative integers a, b, c and partitions λ and μ , there holds

$$\begin{aligned}
 & \text{CT}_x x_0^{|\lambda|-|\mu|} P_\lambda(x^{-1}; q, q^c) P_\mu\left(\left[\frac{q^{c-b-1} - q^a}{1 - q^c} x_0 + \sum_{i=1}^n x_i\right]; q, q^c\right) \\
 & \quad \times \prod_{i=1}^n (x_0/x_i)_a (qx_i/x_0)_b \prod_{1 \leq i < j \leq n} (x_i/x_j)_c (qx_j/x_i)_c \\
 & = (-1)^{|\lambda|} q^{\sum_{i=1}^n (\lambda_i + 1) - cn(\lambda)} P_\lambda\left(\left[\frac{1 - q^{nc}}{1 - q^c}\right]; q, q^c\right) P_\mu\left(\left[\frac{q^{c-b-1} - q^{a+nc}}{1 - q^c}\right]; q, q^c\right) \\
 & \quad \times \prod_{i=1}^n \prod_{j=1}^{\ell(\mu)} (q^{b+(i-j-1)c - \lambda_i + \mu_{j+1} + 1})_{\mu_j - \mu_{j+1}} \prod_{i=1}^n \frac{(q^{a+(n-i)c + \lambda_i + 1})_{b - \lambda_i} (q)_{ic}}{(q)_{b+(i-1)c - \lambda_i + \mu_1} (q)_c}.
 \end{aligned}$$

The above identity reduces to the q -Morris identity when $\lambda = \mu = \emptyset$.

The AFLT-type q -Baker–Forrester constant term identity

Let n, m be positive integers such that $m \leq n$, denote

$$X_m := x_1 + \cdots + x_m, \quad X_{n \setminus m} := x_{m+1} + \cdots + x_n$$

and

$$X_m^{-1} := x_1^{-1} + \cdots + x_m^{-1}, \quad X_{n \setminus m}^{-1} := x_{m+1}^{-1} + \cdots + x_n^{-1}$$

in **plethystic notation**.

The AFLT-type q -Baker–Forrester constant term identity

For positive integers n, m such that $m \leq n$, and nonnegative integers a, b, c , denote $F_{n,m}(x; a, b, c)$ to be

$$\prod_{i=1}^n (x_0/x_i)_a (qx_i/x_0)_b \prod_{1 \leq i < j \leq n} (x_i/x_j)_{c+\chi(j \leq m)} (qx_j/x_i)_{c+\chi(j \leq m)}.$$

For partitions λ, μ, ξ such that $\ell(\lambda), \ell(\mu) < m$ and $\ell(\xi) \leq n - m$, denote $F_{n,m}(a, b, c, \lambda, \xi, \mu)$ to be

$$\begin{aligned} & \text{CT}_x x_0^{|\lambda|+|\xi|-|\mu|} F_{n,m}(x; a, b, c) P_\xi \left(X_{n \setminus m}^{-1}; q^{c+1}, q^c \right) \\ & \times P_\lambda \left(\left[X_m^{-1} + \frac{1 - q^c}{1 - q^{c+1}} X_{n \setminus m}^{-1} \right]; q, q^{c+1} \right) \\ & \times P_\mu \left(\left[\frac{q^{c-b} - q^a}{1 - q^{c+1}} x_0 + X_m + \frac{q - q^{c+1}}{1 - q^{c+1}} X_{n \setminus m} \right]; q, q^{c+1} \right). \end{aligned}$$

Theorem 1.1. Let $F_{n,m}(a, b, c, \lambda, \xi, \mu)$ be defined in (1.7). If λ and ξ satisfy at least one of the three conditions:

- (i) $\xi = \emptyset$; (ii) $\lambda = \emptyset$ and $\xi_1 \leq m$; (iii) $\lambda, \xi \neq \emptyset$, and $|\lambda| + |\xi| \leq \min\{n - m, m\}$,

then

$$\begin{aligned}
 (1.8) \quad & F_{n,m}(a, b, c, \lambda, \xi, \mu) \\
 &= (-1)^{|\mu| + \sum_{i=m+1}^n \eta_i} q^{(b+1)(|\xi| + |\lambda|) + \sum_{i=1}^{\ell(\mu)} \binom{\mu_i}{2} - (c+1)n(\mu) - \sum_{i=1}^{n-m} ((ic+b)\eta_{i+m}^+ - \binom{\eta_i^+ + m}{2})} \\
 &\times P_{\xi}^{(q^{c+1}, q^c)} \left[\frac{1 - q^{(n-m)c}}{1 - q^c} \right] P_{\lambda}^{(q, q^{c+1})} \left[\frac{1 - q^{nc+m}}{1 - q^{c+1}} \right] P_{\mu}^{(q, q^{c+1})} \left[\frac{1 - q^{b+(n-1)c+m+a}}{1 - q^{c+1}} \right] \\
 &\times \prod_{i=1}^{m-1} \prod_{j=1}^{m-1} (q^{(i-j)(c+1) - b - \mu_i + \lambda_{j+1}})_{\lambda_j - \lambda_{j+1}} \prod_{i=1}^{m-1} \frac{(1 - q^{(i+1)(c+1)})(q)_{ic-b-1+\eta_{m-i}^+}}{(q)_{i(c+1)-b-1-\mu_i+\lambda_1}} \\
 &\times \prod_{i=1}^n \frac{(q^{(n-i)c+1+\eta_i^+ + a})_{b-\eta_i^+ + \chi(i \leq m)(m-i)} (q)_{ic+\chi(i > n-m+1)(i-(n-m+1))}}{(q)_{(i-1)c+b-\eta_{i+m-1}^+ + \chi(i \geq n-m+1)(i-(n-m+1)+\mu_{n-i+1})} (q)_{c+\chi(i > n-m+1)}},
 \end{aligned}$$

Here $\eta := (\lambda + \delta^m, \xi)$, $\delta^m := (m-1, \dots, 1, 0)$, and η^+ is the unique partition obtained by rearranging the entries of η in decreasing order.

End! Thanks!